# Hippolyte Mayard

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GitHub LinkedIn Google Scholar Personal webpage


## Senior ML Scientist - Medical Imaging & Computer Vision | Research to Production

6+ years shipping computer-vision models to regulated production: 2D/3D medical imaging, two FDA 510(k) submissions, false-positive and operating-point control for AI used by 50,000+ clinicians in 23,000+ practices.

## PROFESSIONAL EXPERIENCE

**2025** — Senior ML Scientist at Pearl AI, *Remote (Italy) - US dental AI, FDA-cleared 2D + 3D imaging*

*Jan-present*

- Technical owner of the 3D CBCT pathology program (5 pathologies, 8 anatomical structures): lead since Feb 2026, 2 FDA 510(k) submissions filed, Second Opinion 3D shipped commercially in 7 months.
- 2D caries detection on intra-oral and panoramic X-rays, used by 50,000+ clinicians in 120+ countries: cut the false-positive rate driving churn at constant recall; shipped caries progression on top.
- Operating-point design for real users: clinician personas (recall-first vs specificity-first), default threshold set between them, per-site sensitivity presets with per-class PR curves, per-device robustness checks.
- Production inference on AWS, versioned algorithm packages, MLflow tracking, GitLab CI.
- Regulatory evidence for two 510(k)s: validation protocol, locked test set with enforced train/test independence, performance reports, technical file; 3D findings turned into claim-ready 2D documentation.

*Stack: Python, PyTorch, MONAI, AWS, Docker, MLflow, FastAPI, GitLab CI*

**2026** — Freelance Software Engineer - Clinica Linfologica Italiana, *Italy (remote)*

*May-Aug*

- Solo contractor: patient booking PWA for a lymphology clinic (Next.js, Supabase, Stripe, Vercel), GDPR-compliant, live in production with online payments and automated reminders.

**2024-2025** — Lead AI Engineer at Theodo Data & AI, *Paris, France*

*Aug-Jan*

- Surgitec-Robotics: 3D brain vessel segmentation for surgical trajectory planning, minimally invasive surgery.

**2021-2024** — Founding Deep Learning Engineer at Allisone.ai, *Paris, France*

*Nov-Jul*

- Built and deployed deep-learning models for classification, detection and segmentation on 2D panoramic and intra-oral dental X-rays, powering clinical diagnostic features.
- Designed and implemented advanced AI algorithms for detection, segmentation, and generation tasks on 3D Cone Beam Computed Tomography (CBCT) data.

*Stack: Python, PyTorch, Git, AWS, Docker, FastAPI, ClearML*

**2020-2021** — Research Engineer at INRIA (team Epione), *Sophia Antipolis, France*

*Apr-Nov*

Metal Artefact Reduction on Cone Beam CT scans, under the supervision of Hervé Delingette

*Stack: Python, PyTorch, Git*

**2019-2020** — Data Scientist at Société Générale, Data Lab GTPS, *Paris La Défense*

FX margin analysis, transaction data

## PUBLICATIONS

**2024** — SISTR: Sinus and Inferior alveolar nerve Segmentation with Targeted Refinement on Cone Beam Computed Tomography images

*Laura Misrachi, Emma Covili, Hippolyte Mayard, Christian Alaka, Jérémy Rousseau, Willy Au* - medRxiv

**2023** — Pre-Training with Diffusion Models for Dental Radiography Segmentation - MICCAI 2023, DGM workshop

*Jérémy Rousseau, Christian Alaka, Emma Covili, Hippolyte Mayard, Laura Misrachi, Willy Au* - arXiv

## TEACHING

**2022-2026** — External Lecturer at Université Paris Dauphine-PSL

- Deep Learning for Image Analysis - Master IASD/BDIA - Dauphine - PSL University, *Paris/Tunis*
- Deep Learning - Master BDIA - Dauphine - PSL University, *Tunis*

## EDUCATION

**2020-2021** — Master's degree, MVA (Mathematics, Vision, Learning) - ENS Paris-Saclay

**2019-2020** — Master's in Artificial Intelligence and Data Science (IASD) - PSL (Dauphine PSL, ENS, Mines ParisTech)

**2014-2018** — Bachelor's degree in Applied Mathematics (with honours) - Université Paris-Dauphine PSL

## LANGUAGES AND COMPUTER SKILLS

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|---|---|
| Programming | Python, R, C, C++, Java, SQL |
| Frameworks | PyTorch, Tensorboard, TensorFlow, Keras, Scikit-Learn, Unit Testing (pytest, unittest) |
| Vision | Torchvision, OpenCV, Pillow, Imgaug, Albumentations |
| Medical frameworks | MONAI, Torchio, NiBabel, Pydicom, SimpleITK; FDA clearance documentation & validation |
| Tools & infra | Git, GitLab CI, Docker, AWS, MLflow, ClearML, FastAPI |
| Languages | French (Native), English (Fluent), Italian (Fluent), German (Elementary) |